

DZERZHINSKOYE, Russian Federation

WMO: 294810

Lat: **56.828N**Lon: **95.232E**Elev: **188**StdP: **99.09**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-38.5	-43.8	0.1	-41.4	-40.5	0.1	-38.0	8.6	-9.4	6.8	-8.3	0.2	150	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.9	19.5	27.9	18.6	26.1	17.7	20.9	26.8	19.9	25.7	18.9	24.5	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	23.4	17.9	13.2	22.6	16.8	12.2	21.4	61.8	26.8	58.1	25.6	54.8	24.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.5	4.8	DB	-43.1	33.1	4.3	1.8	-46.2	34.4	-48.7	35.5	-51.1	36.6	-54.2	37.9	(4)
(5)			WB	-42.2	22.6	3.9	1.5	-44.9	23.7	-47.2	24.5	-49.4	25.4	-52.2	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-20.7	-16.9	-7.0	1.8	9.0	16.8	19.0	15.8	8.4	0.6	-10.4	-17.7
	DBStd	15.18	10.17	8.68	6.80	4.92	5.23	4.11	3.08	3.70	4.30	5.61	8.76	9.82
	HDD10.0	4415	950	752	528	249	83	3	0	3	81	293	612	860
	HDD18.3	6798	1208	986	786	497	291	76	30	94	299	551	862	1118
	CDD10.0	758	0	0	0	3	54	206	277	184	33	1	0	0
	CDD18.3	98	0	0	0	0	3	29	49	17	1	0	0	0
	CDH23.3	1168	0	0	0	2	58	426	477	191	15	0	0	0
	CDH26.7	313	0	0	0	0	11	127	124	49	2	0	0	0
Wind	WSAvg	1.9	1.6	1.7	2.1	2.5	2.5	1.9	1.7	1.6	1.8	2.0	2.0	1.7
Precipitation	PrecAvg	375	14	12	13	18	36	51	59	50	46	28	28	21
	PrecMax	480	33	27	27	34	81	107	162	121	136	51	50	45
	PrecMin	292	5	3	1	3	10	12	11	18	14	9	15	4
	PrecStd	51	7	5	7	9	19	24	28	23	24	10	10	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.8	2.2	8.6	19.8	28.1	32.2	32.0	30.7	25.3	15.9	7.9	1.7
		MCWB	-1.3	-1.0	3.5	10.3	16.1	19.4	20.4	19.7	16.5	9.9	5.2	-0.4
	2%	DB	-1.8	0.1	6.1	15.8	25.0	29.9	29.7	27.9	21.4	12.8	4.8	-0.8
		MCWB	-3.2	-2.0	2.2	8.3	14.4	18.9	20.0	18.8	14.6	8.6	2.5	-2.3
	5%	DB	-4.0	-1.9	4.2	12.7	22.0	27.8	27.8	25.3	18.4	10.4	2.7	-2.7
		MCWB	-5.3	-3.7	0.9	6.9	12.8	18.1	19.2	17.9	13.1	7.2	0.9	-3.9
	10%	DB	-6.5	-4.2	2.4	10.0	19.2	25.7	26.1	23.3	15.6	8.2	0.9	-4.8
		MCWB	-7.5	-5.6	-0.2	5.5	11.7	17.1	18.4	16.8	11.7	5.6	-0.5	-6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-0.4	4.1	11.5	17.7	21.5	22.6	21.5	17.6	10.9	5.4	-0.2
		MCDB	0.5	1.6	7.5	18.7	25.5	28.1	28.6	27.9	23.1	14.1	7.3	1.7
	2%	WB	-3.1	-1.9	2.4	9.1	15.4	20.2	21.3	19.8	15.2	9.1	2.6	-2.1
		MCDB	-1.7	0.1	5.4	14.2	22.8	27.0	26.6	25.8	20.1	12.2	4.3	-1.0
	5%	WB	-5.1	-3.6	1.2	7.4	13.7	19.1	20.4	18.7	13.6	7.5	1.0	-3.9
		MCDB	-4.0	-1.9	3.8	12.2	21.0	25.9	25.6	24.0	17.3	10.3	2.5	-2.7
	10%	WB	-7.4	-5.7	-0.1	5.5	12.0	18.0	19.4	17.4	12.3	5.7	-0.5	-6.0
		MCDB	-6.5	-4.2	2.3	10.1	18.2	24.2	24.6	22.1	15.4	8.1	1.0	-4.8
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.1	11.6	10.1	11.9	12.1	10.8	10.8	9.8	7.8	8.0	8.3
		MCDBR	7.6	9.7	11.9	15.4	16.4	16.7	14.4	14.6	15.8	12.0	7.3	8.9
		MCWBR	6.7	8.1	8.8	9.2	8.2	7.0	6.0	6.8	8.8	8.0	5.8	7.9
	5% WB	MCDBR	7.4	9.4	10.9	14.3	14.4	13.7	11.4	13.2	13.4	11.5	6.9	8.8
		MCWBR	6.7	8.0	8.4	9.0	7.9	6.4	5.7	6.7	7.9	7.9	5.6	8.0
Clear-Sky Solar Irradiance	taub	0.282	0.314	0.354	0.386	0.408	0.424	0.440	0.411	0.365	0.344	0.308	0.276	
	taud	1.998	1.994	1.973	2.139	2.206	2.245	2.222	2.273	2.360	2.255	2.061	1.959	
	Ebn at Noon	552	700	774	821	828	818	794	792	779	677	514	436	
	Edh at Noon	64	98	131	130	130	127	128	114	90	76	60	50	
All-Sky Solar Radiation	RadAvg	0.73	1.73	3.11	4.39	5.05	5.96	5.49	4.40	2.70	1.40	0.69	0.43	
	RadStd	0.08	0.11	0.18	0.26	0.47	0.52	0.35	0.35	0.35	0.11	0.11	0.06	